

James B. Kane

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EDUCATION

Florida Atlantic University

Bachelor of Science in Electrical Engineering
Minor in Computer Science

Boca Raton, FL

Expected Graduation: December 2028

GPA: 3.7

SKILLS

Software: Excel, Solidworks, Desmos, C, C++, C#, Python, LaTeX, Git, VHDL, MATLAB

Hardware: Multimeter, Oscilloscope, FPGAs (Artix-7), Microcontrollers (MSP430, ESP32, Arduino), Soldering

EXPERIENCE

HID Global

Engineering Intern

Fort Lauderdale, FL

May 2026 – Present (Expected End: August 2026)

- Automating validation of a pivCLASS RPKCL40-P reader by designing, constructing, soldering, and programming a custom electromechanical device, resulting in a reliable & hands-free validation system capable of testing up to 120 card credentials per hour without physical human input.
- Orchestrating various hardware modules including 12 push-pull solenoids, a YubiKey PIV smart card emulator, and a color sensor through custom C# and Python programs on a Raspberry Pi 3 & Pi Pico, leading to synchronized, automated interactions with the access reader and a complete card test in 30 seconds.
- Developing a custom housing for the hardware by creating iterative designs in SolidWorks and then 3-D printing, ensuring secure component mounting and structural alignment with tolerances of less than 1.5 millimeters.

Florida Atlantic University

Student Researcher - Team Leader (Vertically Integrated Projects Program)

Boca Raton, FL

January 2026 – Present

- Designing an air quality sensor node to gather data for a predictive model using an MQ-137 Ammonia gas sensor module, ESP32 microcontroller, and C++.
- Collaborating within a multidisciplinary team to deploy and monitor environmental sensor technologies.

Florida Atlantic University

Learning Assistant (General Physics 1 & Calculus 2)

Boca Raton, FL

August 2025 – May 2026

- Lead student groups through complex physics and calculus problems using structured diagramming and equation application, directly improving student scores by an average of 30%
- Collaborated with faculty to implement active-learning strategies for a cohort of 80+ students.
- Served as a technical subject-matter resource, reinforcing core engineering concepts while mentoring freshman teams in collaborative problem-solving.

CLUBS & ORGANIZATIONS

Owl's Racing (Formula SAE) | *Electrical Subsystem Member*

June 2025 – Present

- Assisted in the assembly and troubleshooting of failures for the car's wiring harness.

IEEE-HKN | *Recording Secretary*

April 2026 – Present